

PaySense: AI-Driven Workforce and Payroll Intelligence Platform

Abstract

In modern organizations, Human Resource and Payroll systems play a critical role in managing employee information, attendance, salary computation, and compliance. However, most traditional payroll systems are rule-based and limited to basic salary calculations, lacking the capability to analyze workforce behavior, detect anomalies, or support strategic decision-making. This results in delayed insights, payroll discrepancies, and inefficient human resource management. PaySense is proposed to overcome these limitations by integrating Artificial Intelligence into the HR and payroll lifecycle. The system collects and processes employee data such as attendance records, leave history, salary components, and performance indicators. Machine learning models are employed to analyze attendance patterns, predict leave and attrition risks, forecast payroll costs, and detect salary anomalies or irregularities. Unlike conventional systems, PaySense continuously learns from historical data, enabling adaptive and intelligent payroll management. The platform provides automated payroll computation along with AI-powered insights through an interactive dashboard for HR administrators. By transforming raw payroll data into actionable intelligence, PaySense improves accuracy, transparency, and efficiency in workforce management. The proposed system is designed using a modern web-based architecture with scalable backend services and AI models, making it suitable for real-world organizational deployment.

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